# Low-Latency Joint Sensing Architecture for Bionic Hands Using Piezo Sensor
# Category: Robotics and Intelligent Machines


# Abstract

Bionic hands are essential for restoring grip and dexterity, but their response time is limited by two separate sources of delay. The motor itself takes approximately 200 milliseconds to respond, and this is a fixed mechanical limit. On the other side, ADC sampling and digital filtering add another 80–150 milliseconds before the motor even starts to move. While the mechanical delay cannot be reduced easily, the software - processing delay can be eliminated. This project aims to reduce that delay using a purely hardware-based signal path.

We implemented a piezoelectric disc sensor to detect touch and impact force. Instead of routing this signal through the microcontroller's software for processing, we passed it through an LM393 analog comparator, which directly triggers a microcontroller hardware interrupt, eliminating the software polling loop entirely. The electronic decision-path latency was calculated using manufacturer datasheet values (LM393 propagation delay: 1.3 microseconds; microcontroller interrupt response: 0.3–1 microseconds). This estimate was experimentally validated using a hardware testbed. The measured values (~9 microseconds for both modes) were somewhat higher than the theoretical estimate due to real-world execution overhead, but remained consistent with the overall conclusion.

Our calculations show this design's electronic decision-path latency at approximately 2 microseconds, which is significantly faster, by several orders of magnitude, than published response times for existing prosthetic tactile-sensing systems. These systems typically show response times between 20 milliseconds (in ML-based grasp-detection systems) and 100–200 milliseconds (in advanced e-skin/pressure-sensor designs). This result is specific: it does not make the whole hand faster overall, and it does not replace AI-based grip or force-control processing. Instead, it functions as a small, low-cost, low-weight hardware addition built for fast-reaction critical situations, such as sudden slip or impact, operating as a rapid, reflex-like pathway alongside existing intelligent control systems.

# 1. Introduction & Objective

**Problem:** In bionic/prosthetic hands, the response time is slow compared to human hand response time, especially in sudden slip or impact situations. The motor itself takes approximately 200 milliseconds to respond, and this is a fixed mechanical limit. However, the software process (ADC sampling and filtering) additionally takes typically 80–150 milliseconds for the response every time.

**Objective:** This idea aims to remove the software delay in emergency-like situations, such as sudden slip or impact, by using a hardware-based method (piezo sensor + LM393 comparator + hardware interrupt). It will eliminate the software polling loop and work on direct interruption.

# 2. Innovation

We completely remove the software polling loop and add a hardware-based system. In this system, the piezo sensor's signal goes directly to the LM393 comparator, and it triggers a hardware interrupt. So we do not need a software loop for detection, and it is better than many advanced systems (like SmartHand), which use a complex method — high-resolution sensors and CNN/machine learning for object recognition — and are costly and complex. Our system is a simple analog hardware path, and it is only for fast detection (touch/impact/slip).

# 3. Methodology

We assembled the circuit (piezo + LM393 + Arduino + potentiometer), then set the threshold with the help of the potentiometer. Then we compared the data of two modes:

- Mode A (software polling), where the program continuously checks for a signal even when nothing is happening.
- Mode B (hardware interrupt), where the signal directly triggers the microcontroller without any continuous checking loop.
  We measured the time in Arduino using micros(), so, we could compare both modes, and the data was displayed on the Serial Monitor.

**Testing was completed after a component replacement was required due to initial sensor damage during assembly. A controlled digital trigger (a separate Arduino pin) was used instead of manual tapping, to ensure consistent, repeatable measurements for both modes.**

**Figure 1: Signal-flow architecture — from touch/contact to motor response. (Diagram created with AI tool(Claude) for visualization)**

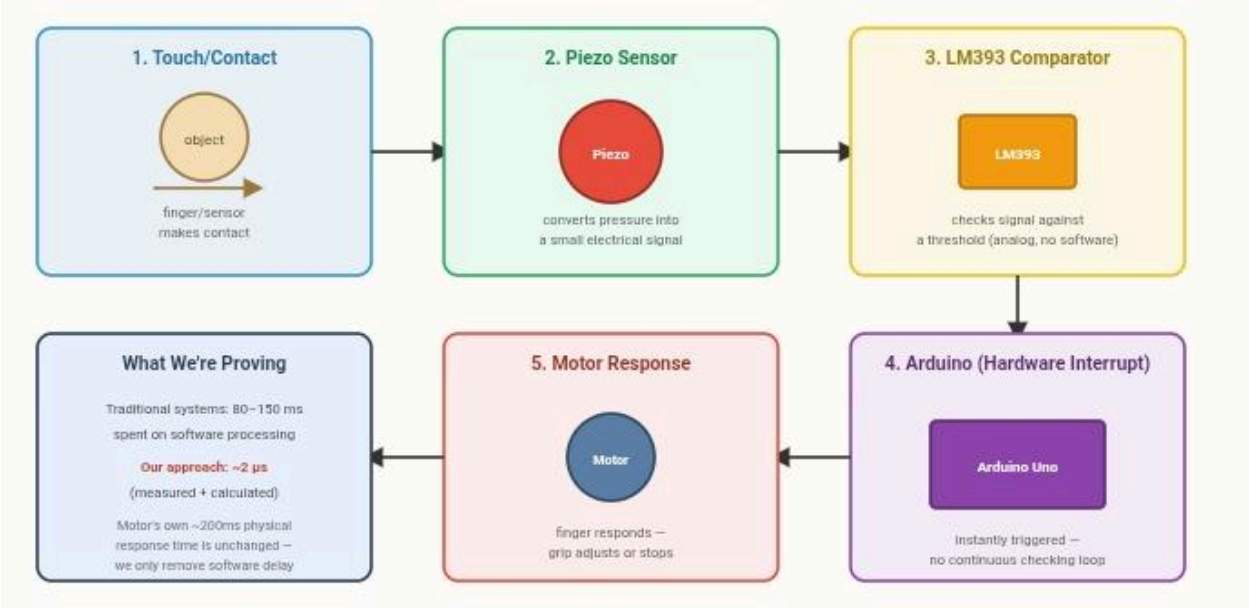
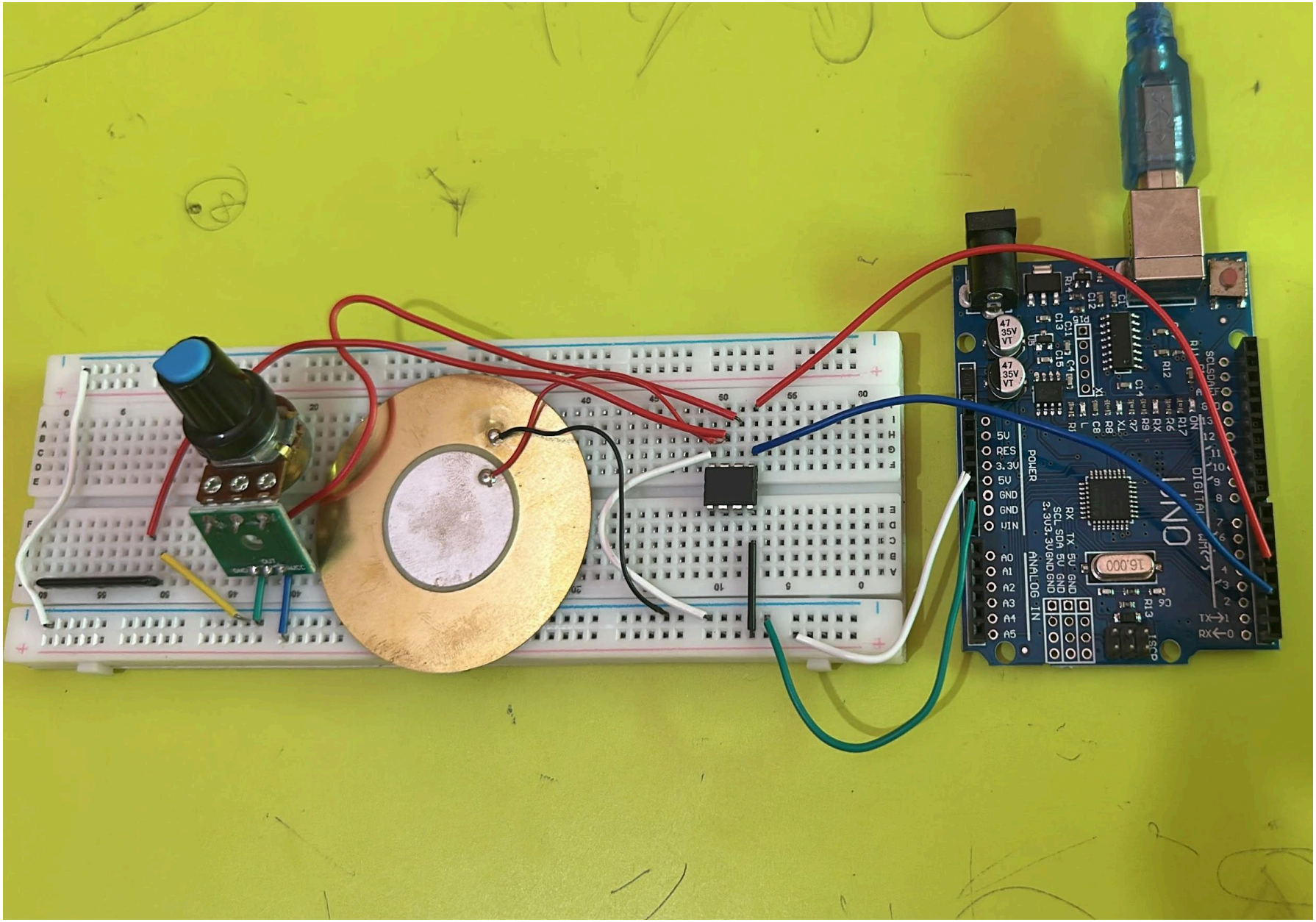



**Table 1: Datasheet-based latency breakdown for the electronic decision-path.**

| Stage | Value | Source |
|---|---|---|
| FPC/wire propagation | Negligible (~ns) | Formula-based |
| LM393 comparator delay | 1.3 µs | Datasheet (TI/ON Semi) |
| MCU interrupt response | 0.3–1 µs | Datasheet (ATmega328) |
| **Total** | **~1.6–2.3 µs** | **Sum** |

**Figure 2: Breadboard testbed used for Mode A/Mode B response-time comparison.**


5V
RES
3.3V
5V
GND
GND
VIN
POWER
A0
A1
A2
A3
A4
A5
ANALOG IN
DIGITAL
16.000
47
35V
VT

## 4. Results & Conclusion

The latency of Mode A (software polling) and Mode B (hardware interrupt) was measured using the hardware testbed, with fifteen trials for each mode. The average measured latency was approximately 9 microseconds for both modes, with individual trials ranging between 8 and 12 microseconds. This is higher than the theoretical estimate (~2 microseconds) due to real-world execution overhead in the microcontroller. Even so, both measured values remain significantly lower than published response times for existing prosthetic tactile-sensing systems, which range from about 20 milliseconds to 100–200 milliseconds.

In this controlled test, both modes produced similar results, likely because the microcontroller was handling a single task with very little overhead. Under a busier workload, Mode A would likely slow down since it must wait its turn to check the sensor, while Mode B would still respond immediately because interrupts happen directly. This suggests Mode B's advantage would become more significant in a real bionic hand, where the microcontroller also has to handle motor control and other processing along with sensor monitoring.

Currently, this is a single-sensor proof of concept. Scaling this approach to a full multi-finger bionic hand will require further work on signal management, wiring, and power efficiency.

## 5. Acknowledgement & References

**Acknowledgement:**

My teammate built and tested the electronic testbed and collected the measurements and My teacher reviewed and signed the approval and safety forms. AI tools were used as follows: Claude, Grok and Gemini for explaining electronics concepts and checking calculations; Claude and Gemini for Arduino test code; Claude and Grok for drafting and language editing of report sections; Claude for literature search and the Figure 1 diagram. The project idea, circuit design, testing and conclusions are our own.

**References:**


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